

INDIANA COUNTY AP, PA, USA

WMO: 725175

Lat: 40.633N

Lon: 79.100W

Elev: 428

StdP: 96.29

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 64706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.0	-21.2	0.6	-12.8	-17.7	0.8	-11.5	11.1	0.1	9.8	-1.1	2.6	260	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.1	23.0	29.1	22.0	27.9	21.3	24.6	28.5	23.8	27.4	23.0	26.1	3.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.4	26.4	22.7	18.4	25.2	22.3	17.9	24.9	77.1	28.5	73.7	27.9	70.7	25.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.1	DB	-21.0	32.6	2.4	2.0	-22.8	34.0	-24.2	35.1	-25.6	36.2	-27.3	37.7	(4)
(5)			WB	-21.1	25.9	2.6	1.2	-23.0	26.8	-24.5	27.5	-25.9	28.2	-27.8	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-2.7	-0.9	3.7	10.4	15.9	20.0	22.1	21.0	18.2	11.7	5.9	0.4	(6)	
	DBStd	9.98	6.77	6.54	6.07	5.34	4.62	3.41	2.79	2.69	3.89	4.94	5.15	5.47	(7)	
	HDD10.0	1467	396	312	212	58	7	0	0	0	1	39	142	300	(8)	
	HDD18.3	3214	652	540	453	243	102	22	4	8	50	214	372	556	(9)	
	CDD10.0	1663	3	5	18	71	191	299	376	342	245	91	20	3	(10)	
	CDD18.3	371	0	0	0	6	28	72	122	91	45	8	0	0	(11)	
	CDH23.3	2775	0	0	2	60	252	552	913	613	333	46	2	0	(12)	
	CDH26.7	699	0	0	0	9	50	149	266	133	86	6	0	0	(13)	
(14) Wind	WSAvg	2.6	3.6	3.4	3.2	3.1	2.4	2.1	1.9	1.6	1.8	2.4	2.9	3.4	(14)	
(15) Precipitation	PrecAvg	1185	105	75	104	107	101	114	126	106	97	79	95	97	(15)	
	PrecMax	1487	215	137	185	178	159	179	230	218	209	156	204	188	(16)	
	PrecMin	934	49	47	48	50	42	47	67	46	32	33	25	59	(17)	
	PrecStd	135	40	22	34	32	34	35	41	50	45	32	41	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.1	22.8	27.8	30.2	32.1	32.7	31.4	31.3	27.4	22.2	17.4	(19)	
		MCWB	13.2	13.7	14.9	18.4	21.4	22.5	23.8	22.8	21.9	21.6	16.4	14.1	(20)	
	2%	DB	13.0	15.1	19.5	25.0	27.8	30.1	31.2	29.1	28.9	24.0	18.9	13.9	(21)	
		MCWB	10.9	11.5	13.3	15.9	19.8	22.4	23.4	22.3	21.7	18.3	14.9	11.4	(22)	
	5%	DB	10.1	12.3	16.3	22.4	26.4	28.6	29.1	27.8	27.3	22.3	17.2	12.0	(23)	
		MCWB	8.1	9.1	11.0	14.5	18.8	21.5	22.4	21.6	20.9	17.5	13.0	10.1	(24)	
	10%	DB	7.3	8.6	13.0	19.7	24.0	27.1	27.8	27.1	25.0	20.0	14.8	8.7	(25)	
		MCWB	5.5	6.1	9.1	13.2	17.9	20.8	21.7	21.1	19.8	16.6	11.4	7.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	14.6	16.6	20.2	23.4	24.8	26.3	25.3	25.0	22.7	18.1	15.0	(27)	
		MCDB	15.6	17.4	20.3	25.6	28.0	29.1	30.4	28.9	28.3	25.9	20.0	16.7	(28)	
	2%	WB	12.1	12.4	14.5	18.1	21.6	23.8	24.9	24.1	23.5	20.7	15.9	12.4	(29)	
		MCDB	13.9	14.5	17.6	21.7	25.8	28.0	29.0	27.6	26.4	22.9	18.1	13.6	(30)	
	5%	WB	8.1	9.2	12.6	16.4	20.4	22.9	23.9	23.3	22.3	18.8	13.8	9.9	(31)	
		MCDB	9.0	11.9	15.7	20.1	24.2	26.6	27.5	26.0	25.0	20.9	16.0	11.4	(32)	
	10%	WB	5.3	6.1	10.0	14.7	19.3	21.9	23.0	22.4	21.2	17.1	11.9	7.0	(33)	
		MCDB	7.1	8.6	13.4	18.4	22.7	25.2	26.2	24.9	23.6	19.1	14.3	8.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.1	10.0	11.8	11.6	10.6	10.5	10.3	10.7	10.1	9.1	7.5	(35)	
		MCDBR	10.3	12.7	13.9	15.4	13.7	12.4	11.9	11.7	12.8	13.1	12.6	11.6	(36)	
	5% WB	MCWBR	8.5	9.8	8.9	8.1	6.7	5.8	5.2	5.5	6.1	7.4	8.6	10.1	(37)	
		MCWBR	9.5	12.1	12.2	12.4	11.4	10.3	10.0	8.9	9.6	10.6	10.8	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.312	0.328	0.354	0.378	0.418	0.459	0.470	0.462	0.381	0.366	0.322	0.303	(40)	
		taud	2.404	2.339	2.355	2.327	2.229	2.144	2.129	2.139	2.348	2.386	2.495	2.484	(41)	
	Ebn at Noon		844	881	896	894	861	824	810	804	855	827	830	830	(42)	
		Edh at Noon	82	101	112	123	139	151	152	147	111	96	75	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.52	2.34	3.31	4.50	5.16	5.64	5.70	5.11	4.17	2.79	1.90	1.25	(44)	
	RadStd		0.13	0.31	0.33	0.43	0.48	0.47	0.38	0.31	0.47	0.29	0.23	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page